

6.9 Multiplying Integers Using Models

Lesson Introduction

 Benchmark	MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.		 Learning Target	I can multiply integers using integer chips, number lines, and patterns.
 Benchmark Coherence	Building On MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.		Working Toward MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- How do you use integer chips and number lines to model multiplication of integers? - What pattern can you use to determine the sign of the product of integers?			
 Lesson Narrative	In this lesson, students use integer chips and number lines to model integer multiplication. Students work with integers that have absolute values less than or equal to 12. The lesson progresses from representing multiplication using integer chips to multiplying integers with opposite signs before closing with multiplying two negative integers. The focus is on students observing a pattern in the signs of the products while making sense of algebraic expressions using models.			
 Component Summary	6.9.1 Warm-Up (~5 min.)	Which One Doesn't Belong? – Number-Line Models (<i>SE p. 274</i>)	6.9.4 Guided Instruction (10 min.)	Discover how to use number lines to model multiplying integers (<i>SE p. 276</i>)
	6.9.2 Exploration (10 min.)	Use integer chips to model multiplication of integers (<i>SE p. 274</i>)	6.9.5 Guided Practice (10 min.)	Multiply integers using integer chips and number lines (<i>SE p. 276</i>)
	6.9.3 Exploration (5 min.)	Explore how zero pairs are used to multiply integers that are both negative (<i>SE p. 275</i>)	6.9.6 Your Turn! (10 min.)	Practice multiplying integers using your preferred method (<i>SE p. 277</i>)
	6.9.7 Wrap-Up (~5 min.)	Determine sign of products of integers (<i>SE p. 277</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Commutative property of multiplication <u>Additional Terms:</u> N/A		- Number lines - Integer chips (Optional) Colored pencils	- If you do not have integer chips, you can print and cut them out from a free online template, have students draw pictorial representations, or use virtual manipulatives. - For 6.9.6 Your Turn! consider having prelabeled blank number lines available.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle multiplying integers if they are not fluent multiplying whole numbers. - Students may not recognize a pattern in the sign of the products when multiplying integers. - Students may struggle to understand the concept of multiplying two negative integers. - Students may not recall the commutative property of multiplication.	- Monitor for students who are correctly modeling the problems with the integer chips and have them share their work with the class. - If you do not have integer chips, students can draw pictorial representations or use virtual integer chips. - Model the precise language of multiplication and how to model it on the number lines accurately in the 6.9.4 Guided Instruction. - If time is an issue, consider assigning 6.9.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Which One Doesn't Belong The 6.9.1 Warm-Up offers a low-floor, high-ceiling opportunity to engage with number lines. This instructional routine allows for a low barrier of entry, as any informal observations (mathematical or not) contribute to the discussion, while students can extend their thinking with a higher ceiling of possible mathematical answers for all four choices.	MA.K12.MTR.3.1: Mathematical Fluency In 6.9.5 Guided Practice, students represent integer multiplication using both integer chips and number lines. Students progress from representing integer multiplication concretely to abstractly as they recognize and internalize the patterns with the signs of the products.
 Differentiate Instruction	If students have delays or deficiencies in their fine motor skills, they can work with a peer tutor or use virtual manipulatives instead of concrete integer chips in the 6.9.6 Your Turn! activities. If students finish 6.9.3 Exploration activities early, ask them to create example multiplication problems with their partners, swap with other pairs, and solve each other's problems.	
Lesson Notes:		

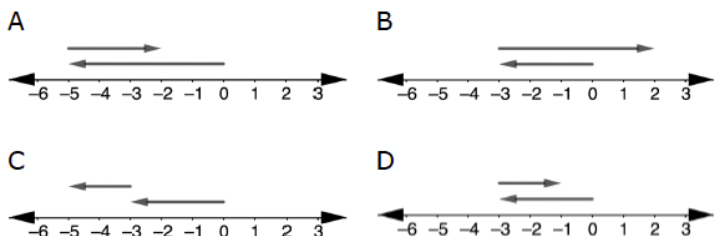
6.9.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students revisit number-line models for adding integers to prepare for multiplying integers on number lines in the lesson. After examining each model, students determine which one they think doesn't belong with the others and justify with mathematics.



6.9.1 Warm-Up: Which One Doesn't Belong?

1. Four number lines modeling addition are shown.



Explain which one doesn't belong.

Answers will vary. C doesn't belong because both arrows are pointing to the left. Every other number line has one arrow starting at 0 and going left, showing a negative number, and one arrow going right, showing adding a positive number. C shows adding two negative numbers.



Which One Doesn't Belong

This instructional routine provides a low-floor, high-ceiling approach to engaging with number lines. Encourage students to choose more than one answer. After they have selected their answer choice and written an explanation, prompt them to find another answer with an explanation of why that one does not belong.



If students struggle, consider providing students with sentence stems such as "I think the _____ doesn't belong because..."

6.9.2 Exploration: Using Integer Chips to Represent Multiplication

Component Overview: Students relate prior knowledge of how to represent multiplication as repeated addition to modeling integer multiplication with integer chips. Students work with integers with the same and with different signs. The commutative property of multiplication is also revisited.



6.9.2 Exploration: Using Integer Chips to Represent Multiplication

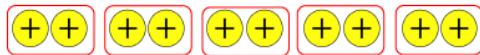
- The addition expression $7 + 7 + 7$ can be written as $3 \cdot 7$. This is because repeated addition of integers can be

expressed as

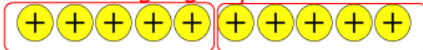
- subtraction.
- **multiplication.**
- division.

- The expression 5×2 can be interpreted as adding 5 groups of 2 or adding 2 groups of 5.
- Use integer chips to represent one of these interpretations.

Recall that $\ominus = -1$ and $\oplus = +1$.
Draw a picture of your representation.
Adding 5 groups of 2



OR Adding 2 groups of 5



- Describe how the integer chips represent the value of the expression 5×2 .
Answers will vary. In the representation, 5×2 is the sum of the 5 groups of 2 positive integer chips. The sum is 10. Likewise, 2×5 is the sum of the 2 groups of 5 positive integer chips. The sum is also 10.
- The expression $5 \times (-2)$ can be interpreted as adding 5 groups of -2 or adding -2 groups of 5.
- Explain which interpretation of $5 \times (-2)$ you think would be easier to represent using integer chips.
Answers will vary. "Adding 5 groups of -2 " is easier to represent using integer chips because this expression means adding 5 groups of 2 negative integer chips. The other interpretation, "adding -2 groups of 5", may be interpreted as the opposite of 2 groups of 5 or $-(2 \times 5)$, which is more difficult to represent using integer chips.



What do you notice about adding 5 groups of 2 and adding 2 groups of 5? Why might this be important?

6.9.2 Exploration: Using Integer Chips to Represent Multiplication (continued)

7. Discuss your choice with your partner. Then, work together to use integer chips to represent one of the interpretations. Draw a picture of the representation.

Answers will vary. 5 groups of -2



OR

-2 groups of 5



8. Describe how the integer chips represent the value of the expression $5 \times (-2)$.

$5 \times (-2)$ means adding 5 groups of negative 2.



The representation shows 10 negative integer chips.

Therefore, the integer chips represent their sum.

OR The other interpretation, "adding -2 groups of 5", may be interpreted as the opposite of 2 groups of 5 or $-(2 \times 5)$. First, (2×5) can be represented as



The opposite of this would then be



9. What is the value of the expression $5 \times (-2)$?

-10



Commutative Property of Multiplication

$$a \times b = b \times a$$

The **commutative property of multiplication** states that the order in which multiplication is performed does not change the value of the expression.

10. Using the commutative property of multiplication, what is the value of the expression -2×5 ?

-10



Encourage mathematical discourse throughout the Exploration, as students have an opportunity to both verbally process and independently or collaboratively write explanations or descriptions in questions 4, 6, 7, and 8. Monitor for student responses that demonstrate understanding or possible misconceptions while students are exploring. Elicit these responses when facilitating a short debrief after the activity.



What do you notice about the value of both expressions?

6.9.3 Exploration: Multiplying a Negative Integer by a Negative Integer

Component Overview: Students explore how zero pairs are used to multiply integers that are both negative. Students learn to think of a negative integer multiplied by a negative integer as removing negative integers from the zero pairs that represent the expression. They generalize about whether the sign will always be positive.



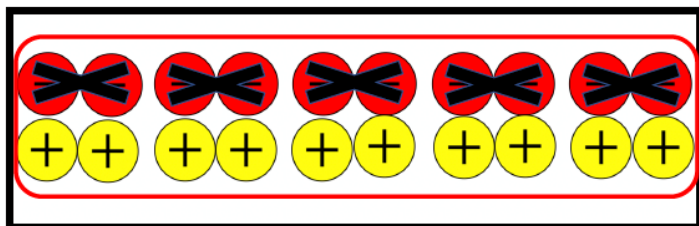
6.9.3 Exploration: Multiplying a Negative Integer by a Negative Integer

1. Complete the statement.

The expression, $(-5)(-2)$, indicates that 5 groups of -2 should be removed.

2. Use the steps below to discover how to represent $(-5)(-2)$ using integer chips.

Step 1: First, use your integer chips to create the set of zero pairs that will be needed to find $(-5)(-2)$. Draw the groups of zero pairs in the box below.



Step 2: Remove 5 groups of 2 negative integer chips.

Step 3: Remove any remaining zero pairs, if needed.

3. Complete the statement.

The value of expression $(-5)(-2)$ is 10.

4. With your partner, discuss whether or not you think the product when multiplying two negative values will always be positive. Explain your thinking.

Answers will vary. I think the product will always be positive because when we create the zero pairs, we are removing groups of negative integers, which only leaves us with positive integers left.



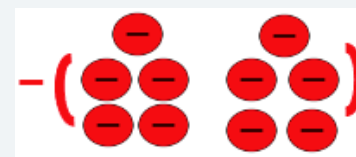
Based on the previous activities, what do you predict the sign will be when multiplying a negative integer by another negative integer?



Challenge students to find more than one way to represent multiplying two negative integers with integer chips accompanied by a different verbal expression, building on the use of the word “opposite” from previous activities.

Option 1: As seen and described in the answer key.

Option 2: “The opposite of two groups of negative five.”



Consider polling the class on question 4. Then ask a few students to justify their responses. After the discussion, repoll the class to see if any students changed their minds.

6.9.4 Guided Instruction: Multiplying Integers Using the Number Line

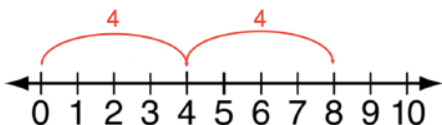
Component Overview: Students are guided through how to model multiplying integers on a number line. Consider having students compare the use of number lines for multiplying integers to using integer chips.



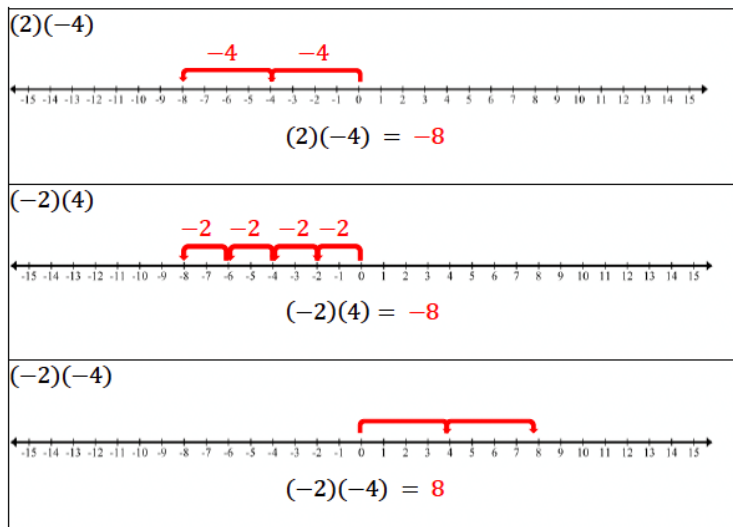
6.9.4 Guided Instruction: Multiplying Integers Using the Number Line

Recall that $(2)(4)$ means 2 groups with 4 units in each group. The product is the total number of objects.

This can be represented on the number line as shown.



1. Represent each multiplication expression on the number line. Then, use it to complete the equation.



Consider displaying the examples, instructions, and visual models from this activity on an anchor chart.



Consider thinking aloud through each of the examples as you demonstrate how to model each expression on the number lines. This provides the opportunity to model precise language and ensure that each expression is modeled correctly on the number line. Consider using the following language for each of the expressions:

$(2)(-4)$ “Two groups of negative 4.”

$(-2)(4)$ “The opposite of two groups of 4.”

$(-2)(-4)$ “The opposite of two groups of negative 4.”

This should be guided instruction, as it will be released to the students in the next component.

6.9.5 Guided Practice: Multiplying Integers Using Models

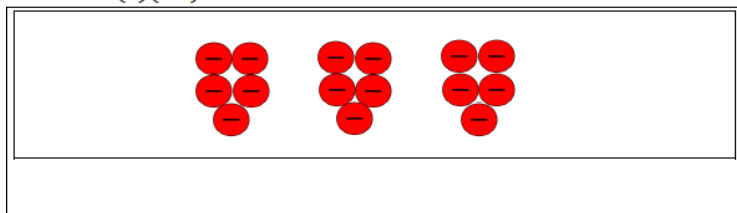
Component Overview: Students model integer multiplication with integer chips and number lines. Students reflect on which representation they prefer to use and why.



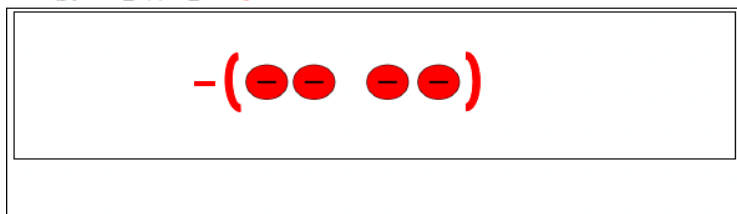
6.9.5 Guided Practice: Multiplying Integers Using Models

- Use integer chips to model and evaluate each expression.

a. $(3)(-5) = -15$



b. $-2 \times -2 = 4$

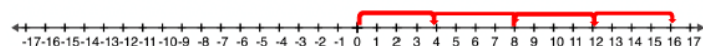


- Use a number line model to evaluate each expression.

a. $(8)(-1) = -8$



b. $(-4)(-4) = 16$



- Which model for multiplication of integers do you prefer: multiplication chips or the number line?

Answers will vary. Number line is faster.



Notice both representations of using integer chips for multiplying integers, with the second representation displaying how students used integer chips in 6.9.2. Prompt students to think creatively and include a different way to represent using integer chips.
Representation 1: “The opposite of two groups of negative two.”
Representation 2: “Two groups of negative two should be removed.”

6.9.6 Your Turn!

Component Overview: Students practice multiplying integers. Instructions include students “using your preferred method” to purposefully highlight and build fluency.



6.9.6 Your Turn!

1. Find the value of each expression using your preferred method.

a. 2×-8
-16

b. $-6 \cdot (-3)$
18

c. $(-7)(-6)$
42

d. $5(-3)$
-15

e. -9×-4
36

f. $7 \cdot 5$
35



Students develop agency over their learning by having the opportunity to choose which representation they prefer to use.



- How did you solve the problem?
- Why did you choose this method?
- Can you solve it using another representation?

6.9.7 Wrap-Up: Cool Down

Component Overview: Students determine the signs of products of integers. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



6.9.7 Wrap-Up: Cool Down

1. Complete each rule by drawing the sign of the product as an integer chip.

Visual Model of Multiplication	Sign of Product
$\oplus \times \ominus$	$\ominus$
$\ominus \times \oplus$	$\ominus$
$\ominus \times \ominus$	$\oplus$
$\oplus \times \oplus$	$\oplus$



Prompt students to provide means of proving their answer. Students can choose to reference an example from the lesson and writing the number next to the assigned row, or students could produce their own examples using integer chips or write a numeric expression.